A Customised Low-voltage Power Supply for a Formula SAE Electric Racing Vehicle

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***Abstract* — This paper presents a custom-built low-voltage (LV) power supply for a Formula SAE (FSAE) electric racing car that eliminates the dependence on the vehicle's LV battery to operate the car's essential functions while the high-voltage (HV) bus is enabled. The power supply consists of a 240 W LLC resonant converter followed by a low-loss priority switch that manages the automatic changeover between a lower-capacity LV battery and the vehicle's HV accumulator battery. The solution incorporates customised circuitry to implement various power converter functions, e.g., start-up supply, under-/over-voltage lockout, over-current protection, and system monitoring/reporting to the vehicle control unit. The overall system (including the priority switch stage) operates at an efficiency of 95 %, and complies with all requirements set by the FSAE rules. Selected experimental results are provided to demonstrate the LV supply operation under dynamic conditions.**

Keywords — Switched-mode power supply, resonant half-bridge converter, Formula SAE, electric racing vehicle.

I. Introduction

Formula SAE (FSAE) is a university-oriented motorsport design competition organised by Society of Automotive Engineers (SAE) International. In the electrical vehicle (EV) category, cars use a 600 V_{dc} high-voltage (HV) battery (accumulator) for propulsion and a separate 12 V_{dc} low-voltage (LV) battery to power essential systems like electronics and cooling [1]. FSAE rules require that these essential systems be energised by the LV battery whilst the HV battery is disconnected. While there are no constraints on the LV battery's capacity, one must balance the benefit of greater capacity and the consequential weight penalty. Challenges with the optimal dimensioning of the LV battery's capacity and the need for separate charging arrangements have led to critical failures during races. As illustrated in Fig. 1, this work proposes to address these issues by replacing the typical higher-capacity LV battery with a galvanically-isolated LV supply derived from the HV battery, that incorporates a smaller LV battery which is only put into operation during the instances the HV battery is disconnected (as per the operational procedures set by the competition rules). This limits the LV battery usage to only a small portion of time until the HV bus becomes available.

The synthesis of the LV supply for an electric vehicle requires a power converter capable of operating at the wide input voltage range available from the car's HV battery and the wide LV output load range that results from the combined operation of ancillary systems such as cooling, monitoring, user interface, and vehicle control. Galvanically-isolated DC-DC converters, such as the phase-shifted full-bridge converter and resonant converter alternatives, are proving popular choices for such applications [2]-[7]. Resonant converters operating with sinusoidal currents offer higher efficiencies at high switching frequencies due to allowing for zero-current (ZCS) or zero-voltage (ZVS) switching capability [6], [7], while the resonant half-bridge LLC variant presents the advantages of a narrower switching frequency range and higher efficiencies across the output load range, compared to its LCC counterpart [7].

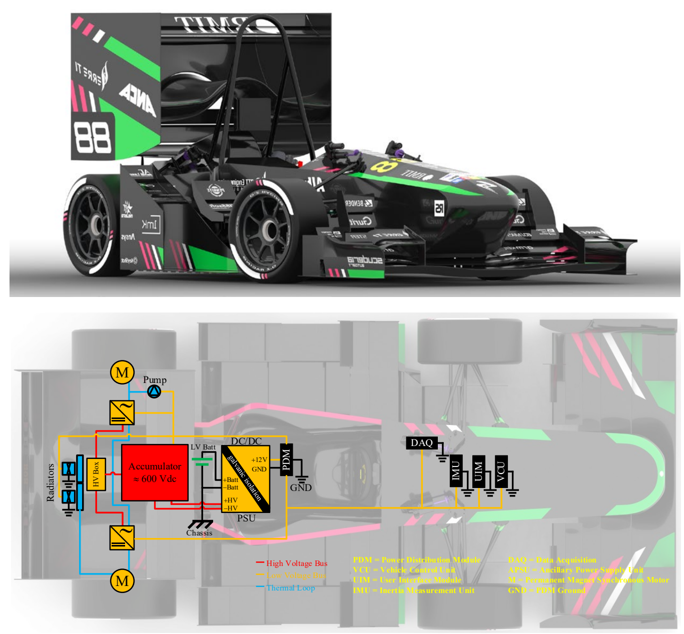

Fig. 1: Overview of a two-wheel-drive Formula SAE electric racing car, illustrating the high-voltage (HV) components and bus (in red) and ancillary low-voltage (LV) components and bus (in orange).

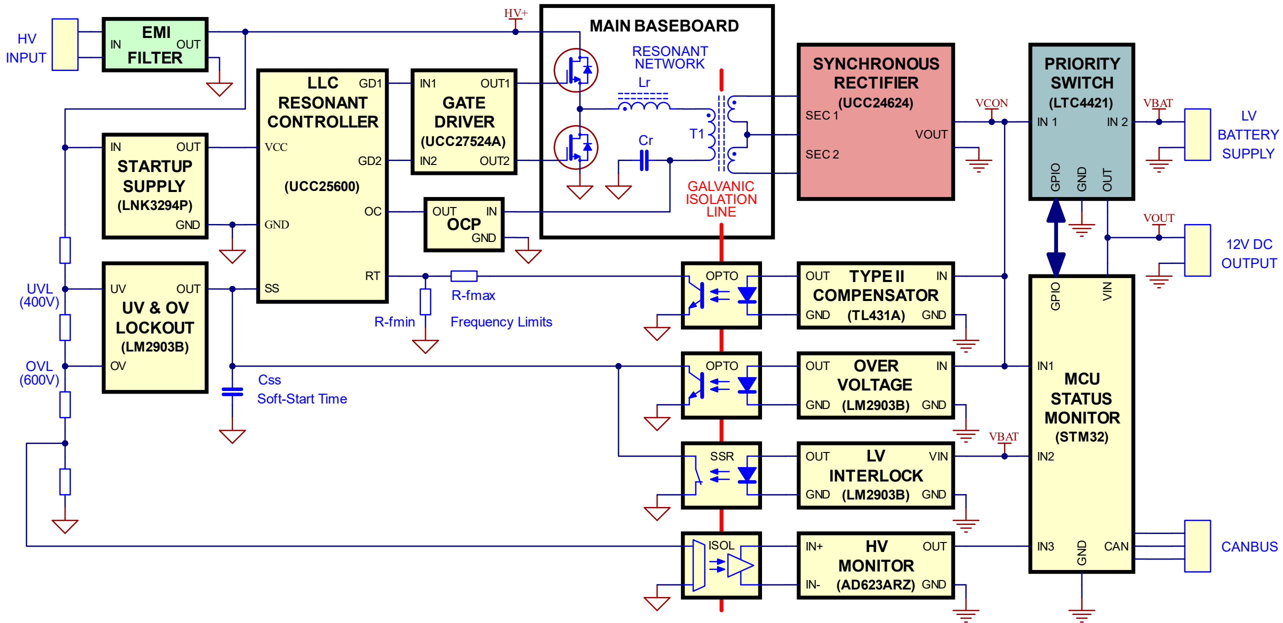

Fig. 2: Block diagram of the low-voltage (LV) power supply including all sub-systems.

II. Overall System Block Diagram

Fig. 2 shows a block diagram of the proposed LV power supply for an electric racing car, built around an LLC-resonant half-bridge power converter, and designed to produce a regulated output voltage of 12 V_{dc} (20 A_{dc} max) from a varying input voltage ranging between 400 V_{dc} and 600 V_{dc}, operating with a variable switching frequency ranging between 100 kHz and 200 kHz, as outlined in Table I.

The function blocks are grouped by colour representing the different PCB modules comprising the overall system. The main baseboard contains the half-bridge devices, resonant network, and high-frequency transformer. The HV input voltage is connected to the half-bridge devices via an input EMI filter. This provides attenuation of unwanted signals in both directions reducing electromagnetic interference. The half-bridge is implemented with 1200 V 70 mΩ Silicon Carbide FETs, and the LLC-resonant power converter is controlled with a Texas Instruments (TI) UCC25600 [8] controller. The system is powered by a non-isolated DC-DC buck converter built using a Power Integrations LNK3294G controller [9] that produces 12 V_{dc} (120 mA_{dc} max) from the incoming HV supply, eliminating the need for an auxiliary transformer winding.

The converter's switching frequency is defined based on the isolated feedback reading of the output voltage. The power switches are driven by a TI UCC27524A dual 5 A low-side gate driver with built-in galvanic isolation.

The half-bridge outputs a square wave with a 50 % duty-cycle and a switching frequency determined by the controller, which feeds the LLC-resonant series network consisting of an external inductor, the transformer's primary-side leakage

TABLE I: System Design Specifications

$V_{in\,(min)}$	$V_{in\,(max)}$	$V_{out\,(nom)}$	$I_{out\,(max)}$	$P_{out\,(max)}$	f_{sw}
400 V_{dc}	600 V_{dc}	12 V_{dc}	20 A_{dc}	240 W	100-200 kHz

inductance, and a resonant capacitor. The transformer's output is a nearly sinusoidal waveform, which is then rectified through a synchronous rectifier to produce the DC output. A TL431-based type II compensator provides isolated feedback for the output voltage regulation.

The system incorporates LV over-voltage and primary-side over-current protections. An LV interlock uses an IXIS LCB717 solid-state relay [10] to deactivate the converter if the LV battery is absent or its voltage is below a threshold, ensuring compliance with the FSAE rules. This functionality is achieved by pulling the LLC controller's soft-start (SS) pin low, as shown in Fig. 2. The system also incorporates HV input voltage monitoring and under-voltage and over-voltage lockouts, both also achieved by pulling the LLC controller's SS pin low. A priority switch, based on the Analog Devices LTC4421 controller [11], manages the switchover from the LV battery to the LV supply when the HV bus becomes active, and vice-versa.

An STM32-based microcontroller monitors the system and shares diagnostics information with the vehicle control unit (VCU) via CAN bus communication. These diagnostics include continuous monitoring of the HV input voltage, LV output voltage, LV battery voltage, and LLC-resonant converter's temperature.

III. LLC Resonant Converter Design

The LLC-resonant power converter was designed using the first harmonic approximation (FHA) method as outlined in [7], [12]. This approach models the converter's input-to-output voltage transfer function M_g to establish a suitable operating region and define the circuit parameters for a given design, assuming the converter operates at or close to the resonant frequency [7], [12], and can be described as the equivalent circuit shown in Fig. 3(a), where V_{ie} is the fundamental component of the resonant tank input voltage, V_{oe} is the fundamental component of the transformer's primary-side

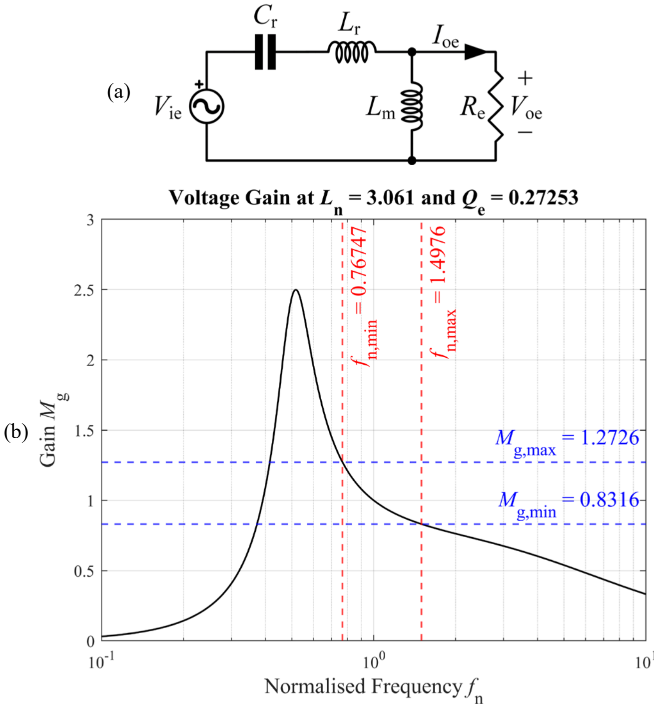

Fig. 3: First harmonic approximation (FHA) method: (a) simplified model of resonant half-bridge converter and (b) MATLAB plot of the input-to-output voltage transfer function M_g for selected values of L_n and Q_e.

voltage, and R_e is the equivalent load resistance. The approach is effective in producing an initial design; however, it requires an iterative process comprising the steps of design, switched simulations (here, Altair PSIM and OrCAD PSpice were used for this purpose), construction, and laboratory testing, to ultimately synthesise an optimised solution.

A MATLAB script was developed to produce a set of plots of the converter's input-to-output voltage transfer function M_g against a normalised frequency $f_n = f_{sw}/f_o$, where f_{sw} is the converter switching frequency and f_o is the resonant frequency, for combinations of the inductance ratio L_n and quality factor Q_e, defined as

$$L_n = \frac{L_m}{L_r} \tag{1}$$

$$Q_e = \frac{\sqrt{L_r / C_r}}{R_e} \tag{2}$$

where L_m is the transformer's magnetising inductance, L_r is the series-resonant inductance, C_r is the series-resonant capacitor, and R_e is the equivalent load resistance, determined from

$$R_e = \frac{8n^2 V_{oe}}{\pi^2 I_{oe}} \tag{3}$$

where n is the transformer's turns ratio. This set of plots (an example of such plots is shown in Fig. 3(b)) was then analysed against the minimum $M_{g,min}$ and maximum $M_{g,max}$ gain requirements, to establish acceptable values for the transformer's magnetising inductance L_m along with the required series-resonant capacitor C_r and series-resonant inductor L_r (which embeds the transformer's primary-side leakage inductance $L_{k,p}$). The need for an iterative process arises from the difficulty in constructing and sourcing components that match the designed parameters, which demands a design-build-test approach until the target performance is achieved.

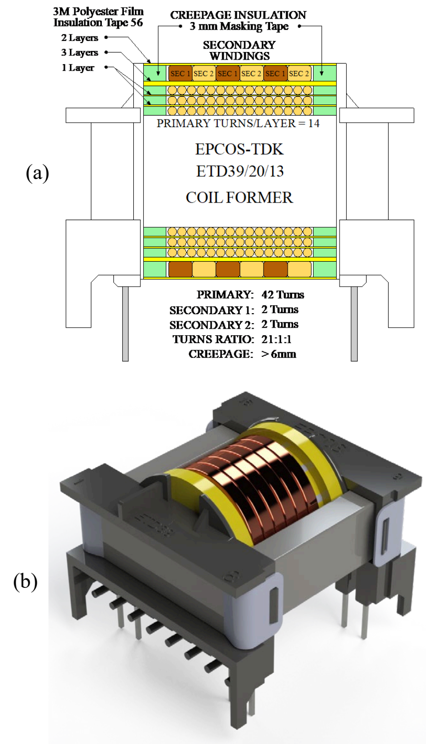

Fig. 4: Customised high-frequency transformer: (a) construction plan and (b) 3D model.

IV. Magnetics Design and Construction

A high-frequency transformer, illustrated in Fig. 4, was iteratively optimised through five successive builds and laboratory measurements using an Omicron Lab Bode 100 Analyser. The design was driven by the system specifications outlined in Table I, which are primarily governed by the application needs, except for the switching frequency range which was a design choice.

The target switching frequency range demanded very thin wires to mitigate the skin effect [13] (which increases AC resistance as the current migrates to the conductor's surface at high frequencies), while the proximity effect [14], [15] (where current concentrates due to nearby magnetic fields) was reduced by twisting wire strands helicoidally [16]. Litz wire was used for both the primary-side and secondary-side windings. The primary winding comprised 160 x 0.07 mm diameter Litz wire, while the secondary winding comprised 1740 x 0.05 mm diameter Litz wire in a flat configuration.

The transformer was constructed with the primary winding wound first on the transformer bobbin. Appropriate creepage and clearance requirements were achieved using two 3 mm insulating barriers made up of masking tape at each end of the bobbin former. This ensured a creepage distance from the outer primary winding of at least 6 mm away from the closest

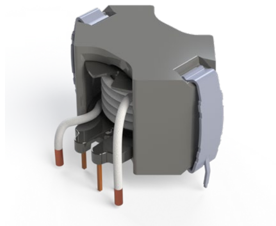

Fig. 5: 3D model of the customised high-frequency external inductor L_r.

secondary conductor. A turns ratio of 21:1:1 was calculated using the system specifications, and a decision was made to use two turns per secondary winding to facilitate fine-tuning, if required, due to typical practical second-order effects on the achievable output voltage. This required a primary winding of 42 turns which fitted neatly over the available bobbin space in 3 layers of 14 turns. A layer of yellow polyester film tape type 56 from 3M was used between each layer of the primary winding to achieve a dielectric breakdown voltage of 4500 V, and three layers of this tape were used between the primary and secondary windings. The secondary windings were wound together to ensure better coupling between the primary and secondary windings as well as reduced leakage inductance. Finally, the transformer was wrapped in another two layers of polyester insulation tape to complete the build. The construction details are provided in Fig. 4(a) and a 3D model of the transformer is shown in Fig. 4(b).

The construction of the external inductor employed the same 3 A 160-strand Litz wire used for the transformer's primary winding. The final iteration had 6 turns wound on a TDK/Epcos RM8 coil former using a 400 nH/turns^2 N87 ferrite core to achieve a measured inductance of 14.56 μH over the entire converter's switching frequency range. A 3D model of the design is shown in Fig. 5.

The resonant network consisting of the inductor, capacitor, and transformer were characterised to establish the resonant frequency of the network. The setup for the measurement using an Omicron Lab Bode 100 Analyser is shown in Fig. 6, where

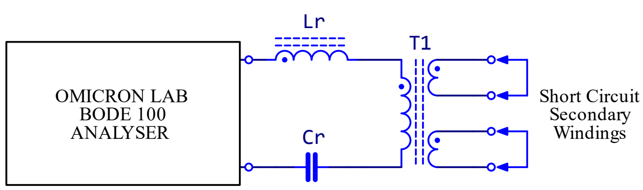

Fig. 6: Test setup for characterisation of the resonant network L_r, C_r, and T1.

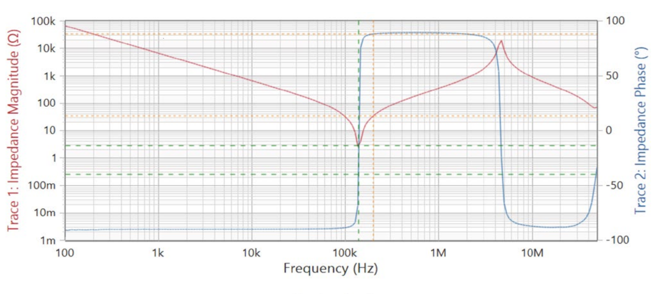

Fig. 7: Impedance measurement of the resonant components L_r, C_r and T1.

(a)

(b)

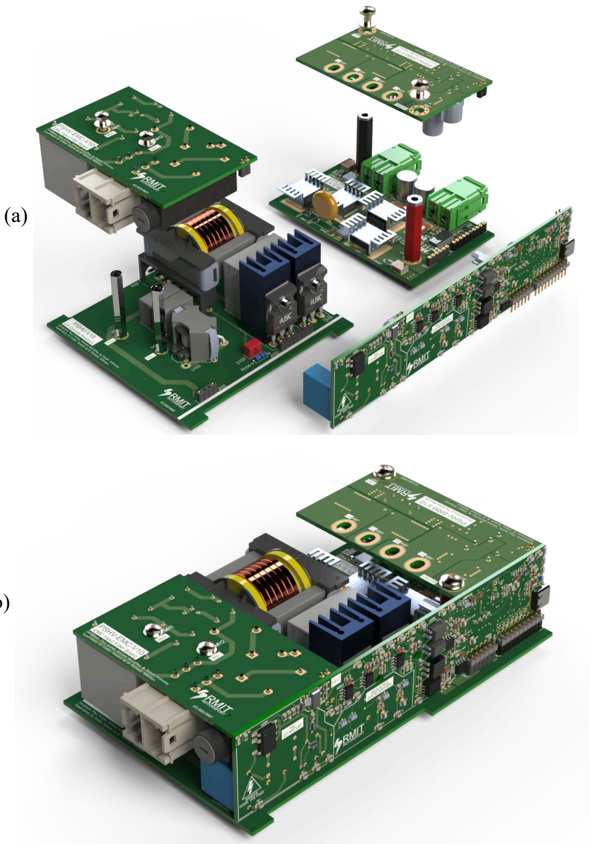

Fig. 8: Modular design approach: (a) exploded view and (b) assembled LV power supply without enclosure.

the three components were series-connected and the transformer secondary-side windings were shorted to ensure that the primary-side leakage inductance was contributing to the measurement. Fig. 7 shows the impedance measured, indicating that resonance occurs at approximately 138 kHz.

V. Modular Design Approach

The development adopted a modular approach for greater flexibility in design and testing and for reducing the overall converter footprint. The solution comprises five modules, and each of them was designed, prototyped, and tested independently before the system integration and commissioning. The PCBs were laid out maintaining loop areas as small as possible by routing supply and return paths directly adjacent to each other on opposite sides of the board. Traces containing high dV/dt signals were kept as short as possible. Safety compliance was achieved using strict creepages and clearances, ensuring the required galvanic isolation was maintained throughout. Fig. 8 shows the design 3D models and how they are assembled together.

VI. Experimental Results

The LV power supply developed in this work was experimentally verified through step-up and step-down load transient events, where the HV input was set to 500 V_{dc}, and the converter LV output was connected to a Chroma 63804 electronic programmable load. Fig. 9(a) shows the load step-up response when the electronic load is commanded to change from

5 Ω to 1.25 Ω with a slew rate of 0.6 A/μs. The figure shows that the primary-side half-bridge switched voltage (red) remains stable throughout the transient, while the load current (black) presents a small overshoot, and the load voltage (blue) remains within the target output voltage range of 12 V_{dc} ± 4%. Conversely, Fig. 9(b) shows the load step-down response when the electronic load is now commanded to change from 1.25 Ω to 5 Ω with the same slew rate of 0.6 A/μs. The figure shows that the primary-side half-bridge switched voltage (red) again remains stable throughout the transient, while the load current (black) presents a small undershoot and the load voltage (blue) stays within the target output voltage range of 12 V_{dc} ± 4%. Measurements using a Hioki PW8001 Power Analyzer provided an overall converter efficiency (including the losses in the priority switch module) of ≈ 95% for the entire operating range of HV input voltage and LV output load.

The ability to dynamically switchover the synthesis of the LV output voltage from the HV battery through the LLC resonant step-down converter operation to the supply of this voltage from the LV battery, and vice-versa, has also been experimentally verified, as illustrated in Fig. 10, where the LV power supply unit was feeding a constant 1.25 Ω output load.

Fig. 10(a) and its enlarged view shown in Fig. 10(b) illustrate the following events: (i) the disconnection of the HV battery and the slow decay of the HV input voltage (green) for $t \in [-200, 0]$ ms, (ii) the DC-DC converter turn-off due to the trigger of its under-voltage lockout (UVL, 400 V_{dc}) at $t \approx -0.5$ ms and the consequential slower decay of the HV input voltage (green) from this time onwards due to the converter turn-off, (iii) the resulting decay of the LV output voltage (black) for $t \in [-0.5, 0]$ ms, (iv) the trigger of the priority switch changeover when the converter output voltage falls below ≈ 9.5 V at t = 0 ms, (v) the sharp increase of the LV battery current (red) from 0 A to ≈ 9.5 A, with some overshoot (and the consequential decrease of the converter output current from ≈ 9.5 A to 0 A, not shown

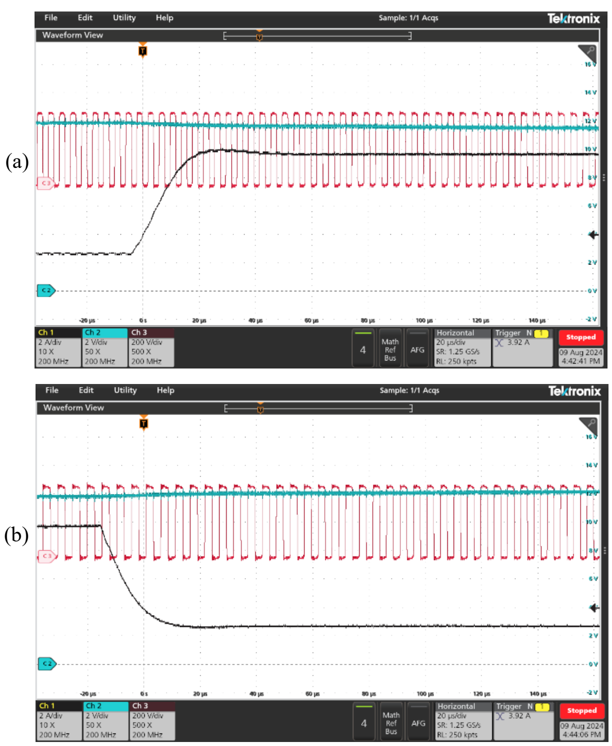

Fig. 9: Experimental load transient response: (a) load step-up and (b) load step-down, showing switched primary-side half-bridge voltage (red, 200 V/div), load voltage (blue, 2 V/div), and load current (black, 2 A/div).

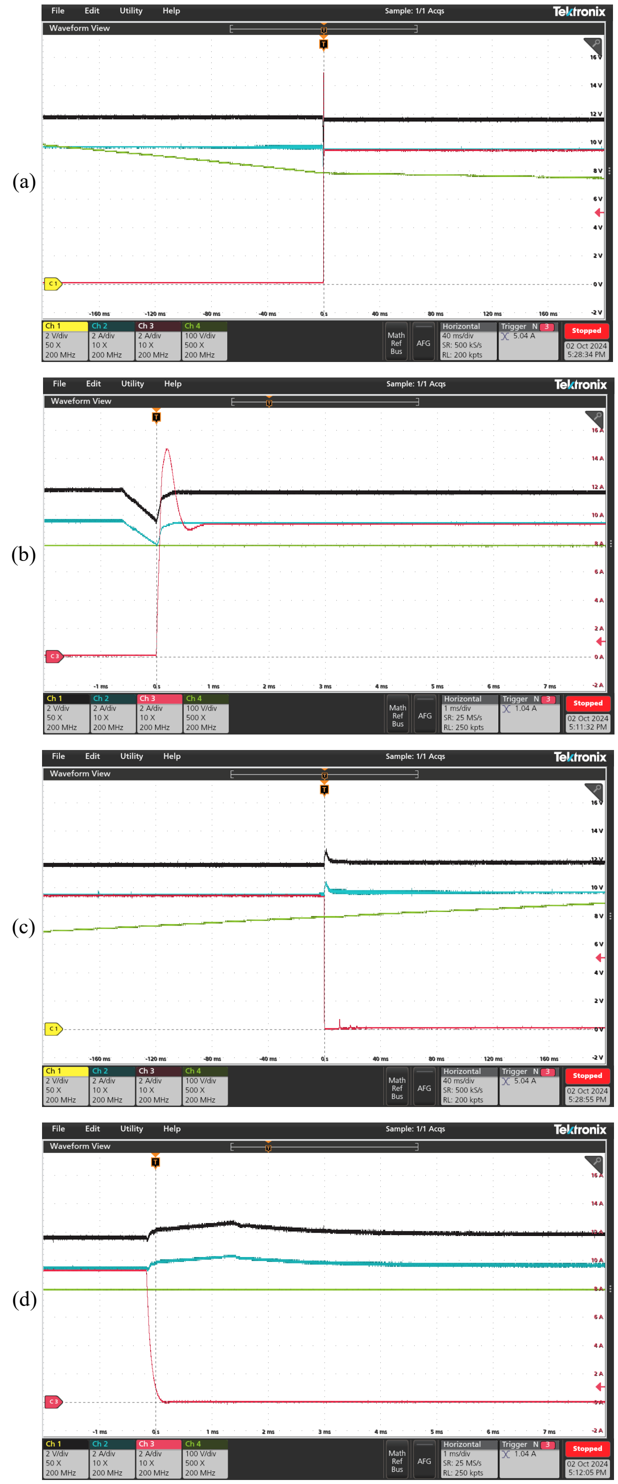

Fig. 10: LV supply changeover events: (a) changeover from converter supply to battery supply, (b) enlarged view of (a), (c) changeover from battery supply to converter supply, and (d) enlarged view of (c). HV input voltage (green, 100 V/div), LV output voltage (black, 2 V/div), LV output current (blue, 2 A/div), and LV battery input current (red, 2 A/div).

in the plots) for t ∈ [0, 1] ms, and (vi) the slight reduction of the LV output voltage (black), due to a smaller LV battery voltage compared to the converter output voltage, for t > 1 ms.

Similarly, Fig. 10(c) and its enlarged view shown in Fig. 10(d) illustrate the following events: (i) the connection of the HV battery and the slow raise of the HV input voltage (green) from t = –200 ms, (ii) the DC-DC converter turn-on due to the release of its under-voltage lockout (UVL, 400 V_{dc}) at t ≈ –0.1 ms, (iii) the trigger of the priority switch changeover when the converter output voltage rises above ≈ 10.7 V at t = –0.1 ms (note that this voltage is internal to the converter and not show in these plots), (iv) the sharp decrease of the LV battery current (red) from ≈ 9.5 A to 0 A (and the consequential increase of the converter output current from 0 A to ≈ 9.5 A, not shown in the plots) for t ∈ [–0.1, 0.1] ms, and (vi) the slight increase of the LV output voltage (black), due to a larger converter output voltage compared to the LV battery voltage, for t > –0.1 ms.

Both dynamic switchover events occur quickly, with the LV output voltage and current reestablished in less than 2 ms.

VII. Conclusion

This paper has presented a customised low-voltage supply system for an FSAE electric racing vehicle that replaces the typically large LV battery with a galvanically isolated DC-DC converter, a substantially smaller LV battery (only used while the HV bus is disconnected), and a priority switch that manages the switchover from the LV battery to the DC-DC converter's regulated output when the HV bus is in operation.

The proposed solution (as shown in Fig. 11) reduces the overall volume and weight of the low-voltage supply system and eliminates the reliance on the LV battery charge for dynamic vehicle's operation, which is known to bring cars to a full stop during competitions if the LV battery dimensioning is inconsistent with the LV system power consumption during endurance events.

The system contains tailored functionalities to comply with the stringent FSAE rules and incorporates diagnostics capabilities, which are continuously communicated to the VCU via CAN bus.

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Fig. 11: Photo of LV power supply packaged in Aluminium enclosure.